



Linux Terminal

General Commands

<code>clear</code>	Clear the terminal screen.
<code>history</code>	Display the command history.
<code>cal</code>	Display a calendar.
<code>uptime</code>	Display how long the system has been running.
<code>man</code>	Display the manual for a command. (Tap q to quit the manual)
<code>date</code>	Display the current date and time.
<code>date "+Y-m-d 00:00:00"</code>	set the date and time.

File Management Commands

<code>cd</code>	Change directory.
<code>cd ..</code>	Move to the previous directory.
<code>cd ~</code>	Move to the home directory.
<code>cd /</code>	Move to the root directory.
<code>ls</code>	List files in the current directory.
<code>ls -l</code>	List files with detailed information.
<code>ls -a</code>	List all files, including hidden ones.
<code>pwd</code>	Print the current working directory.
<code>open</code>	Open a file with the default application.
<code>mkdir</code>	Create a new directory.
<code>rm</code>	Remove a file.
<code>rm -r</code>	Remove a directory and its contents.
<code>cp</code>	Copy a file or directory.
<code>mv</code>	Move or rename a file or directory.
<code>touch</code>	Create a new empty file.
<code>cat</code>	Display the contents of a file.
<code>tail</code>	Display the last few lines of a file.
<code>tail -f</code>	Continuously display new lines added to a file.
<code>head</code>	Display the first few lines of a file.
<code>stat</code>	Display detailed information about a file.

OS Commands

<code>sudo</code>	Execute a command with superuser privileges.
<code>sudo -l</code>	List permitted commands.
<code>whoami</code>	Display the current user.
<code>who</code>	Display all logged-in users.
<code>id</code>	Display user ID and group ID information.
<code>id -u</code>	Display the user ID.
<code>id -g</code>	Display the group ID.
<code>ps</code>	Display currently running processes.
<code>kill</code>	Terminate a process by its PID.
<code>kill -9</code>	Forcefully terminate a process by its PID.
<code>reboot</code>	Restart the system.
<code>poweroff</code>	Shut down the system.
<code>useradd</code>	Add a new user.
<code>userdel</code>	Delete a user.
<code>passwd</code>	Change a user's password.
<code>su - user</code>	Switch to another user.
<code>chown user file</code>	Change the owner of a file to user .
<code>chown user:group file</code>	Change the owner and group of a file.
<code>chmod flag file -u tom</code>	Change the permissions of file for user tom .
<code>chmod +x file</code>	Add execute permission to a file.
<code>chmod +r file</code>	Add read permission to a file.
<code>chmod +w file</code>	Add write permission to a file.
<code>chmod -x file</code>	Remove execute permission from a file.
<code>chmod -r file</code>	Remove read permission from a file.
<code>chmod -w file</code>	Remove write permission from a file.

Daemons

Systemd

Systemd is the most widely used init system and service manager in modern Linux distributions. It is used in distributions such as Ubuntu, Debian, Fedora, and many others.

<code>systemctl start service</code>	Start a service.
<code>systemctl stop service</code>	Stop a service.
<code>systemctl restart service</code>	Restart a service.
<code>systemctl status service</code>	Check the status of a service.
<code>systemctl enable service</code>	Enable a service to start on boot.
<code>systemctl disable service</code>	Disable a service from starting on boot.
<code>systemctl daemon-reload</code>	Reload the systemd manager configuration.
<code>systemctl reset-failed service</code>	Reset the failed state of a service.
<code>journalctl -u service</code>	View logs for a specific service.
<code>journalctl -n 100 -u service</code>	View the last 100 log entries for a specific service.
<code>--no-pager</code>	Disable paging of log output.

OpenRC

OpenRC is an alternative init system and service manager used in some Linux distributions, such as Alpine Linux and Gentoo.

<code>rc-service myService start</code>	Start a service.
<code>rc-service myService stop</code>	Stop a service.
<code>rc-service myService restart</code>	Restart a service.
<code>rc-service myService status</code>	Check the status of a service.
<code>rc-update add myService default</code>	Enable a service to start on boot.
<code>rc-update del myService default</code>	Disable a service from starting on boot.

Shell Scripting

operators

<code>></code>	Redirect output to a file, overwriting the file if it exists.
<code>>></code>	Redirect output to a file, appending to the file if it exists.
<code><</code>	Redirect input from a file.
<code> </code>	Pipe the output of one command to another command.
<code>&&</code>	Execute the next command only if the previous command succeeded.

Basic Syntax

<code>#!/bin/bash</code>	Shebang line to specify the script interpreter.
<code>echo</code>	Print text to the terminal.
<code>read</code>	Read input from the user.
<code>if</code>	Start of a conditional statement.
<code>elif</code>	Else if condition in a conditional statement.
<code>else</code>	Else condition in a conditional statement.
<code>fi</code>	End of a conditional statement.
<code>for</code>	Start of a for loop.
<code>while</code>	Start of a while loop.
<code>do</code>	Start of the loop body.
<code>done</code>	End of a loop.
<code>\$0</code>	The name of the script.
<code>\$1, \$2, ...</code>	Positional parameters for script arguments.
<code>grep</code>	Search for a pattern in a file or input.
<code>exec</code>	Execute a command.
<code>exit</code>	Exit the script.

Networking Commands

The below commands requires the **net-tools** package to be installed on the system.

ifconfig	Displays IP address.
ping	Returns packet loss between two the device and a destination.
traceroute	Traces the path packets take to a destination.
nslookup	Returns IP address information for a domain name.
arp -a	Displays the ARP table, which maps IP addresses to MAC addresses.
arp myIPa	Displays the ARP table for the specific IP address.
curl	Transfers data from or to a server, using various protocols.
curl ifconfig.me	Display machine’s public IPv4 address.

SSH connection

ssh -i ~/.ssh/myKey user@ipv4	Connect to a remote host via SSH.
exit	Exit the SSH session.
Ctrl + D	Exit the SSH session.

Cron

Cron is a time-based job scheduler. It allows users to schedule tasks (commands or scripts) to run automatically at specified intervals.

Basics

crontab -l	List the current user’s crontab.
crontab -e	Edit the current user’s crontab.
crontab -r	Remove the current user’s crontab.
crontab filename	Install a new crontab from a file.
crond	The cron daemon sometime also called cron .

Syntax

* * * * * command	Run command every minute.
0 * * * * command	Run command at the start of every hour.
0 0 * * * command	Run command at midnight every day.
0 0 * * 0 command	Run command at midnight every Sunday.
0 0 1 * * command	Run command at midnight on the first day of every month.
0 0 1 1 * command	Run command at midnight on January 1st every year.

After the time and date fields, you can specify the command to

be executed (note that cron dont use \$PATH so you need to use absolute paths). Example:

```
0 0 * * 0 /usr/bin/python3 /home/user/myscript.py
0 5 * * * /sbin/reboot
```

This will run the Python script **myscript.py** at midnight every Sunday and reboot the system at 5 AM every day.

package managers

apt	Debian-based(include ubuntu) package manager.
apk	Alpine Linux package manager.
yum	Red Hat-based package manager.
dnf	Fedora-based package manager.
pacman	Arch Linux package manager.
brew	Alternative package manager for macOS and many Linux distributions.

apk

update	Update repository indexes.
upgrade	Install upgrades available from repositories .
list	List all available packages.
add pkg	Install a package named pkg .
del pkg	Remove a package named pkg .
search pkg	Search for a package named pkg .
info pkg	Display detailed information about a package named pkg .

apt

update	Update the package list.
upgrade	Upgrade all installed packages to their latest versions.
list	List all available packages.
list --upgradable	List all packages that can be upgraded.
install pkg	Install a package named pkg .
remove pkg	Remove a package named pkg .
search pkg	Search for a package named pkg .